Introduction and Structures of Materials - IIIb



MO-201 Materials Science
Odd Sem 2026-27

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Symmetry Elements

- The **identity** operation is the simplest symmetry element of all – it does nothing! But it is important because all objects at the very least have the identity element, and there are many objects that have no other symmetry elements

- The **reflection** is the symmetry operation that occurs when we put an object in front of a mirror; the image is found perpendicular to the reflection plane and equidistant from that plane, on the opposite side of the plane

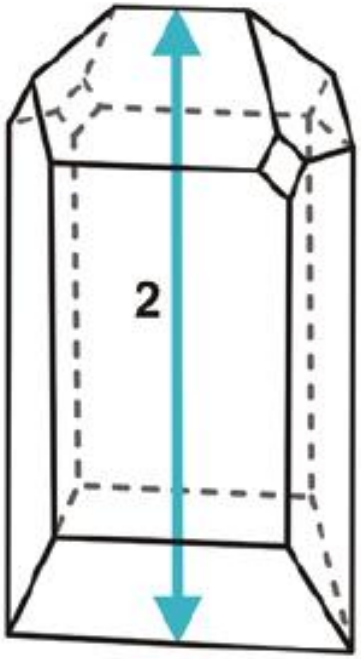
- The **inversion** operation occurs through a single point called the inversion center (center of symmetry); each part of the object is moved along a straight line through the inversion center to a point at an equal distance from the inversion center

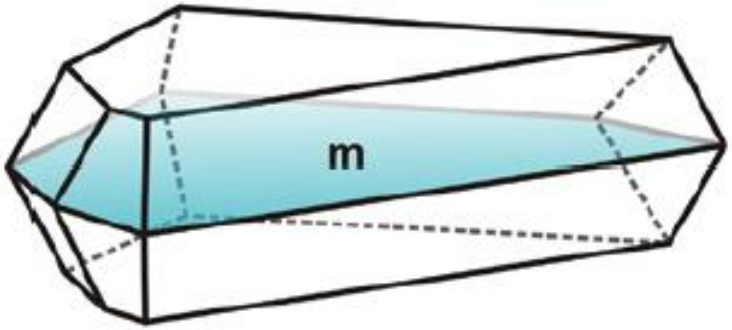
- The **rotation** operations (both proper and improper) occur with respect to a line called rotation axis; a proper rotation is performed by rotating the object 360°/n, where n is the order of the axis; an improper rotation is performed by rotating the object 360°/n followed by a reflection through a plane perpendicular to the rotation axis

- The **rotoinversion** operation involves a rotation plus an inversion

https://www.xtal.iqfr.csic.es/Cristalografia/parte_03-en.html

Symmetry Elements



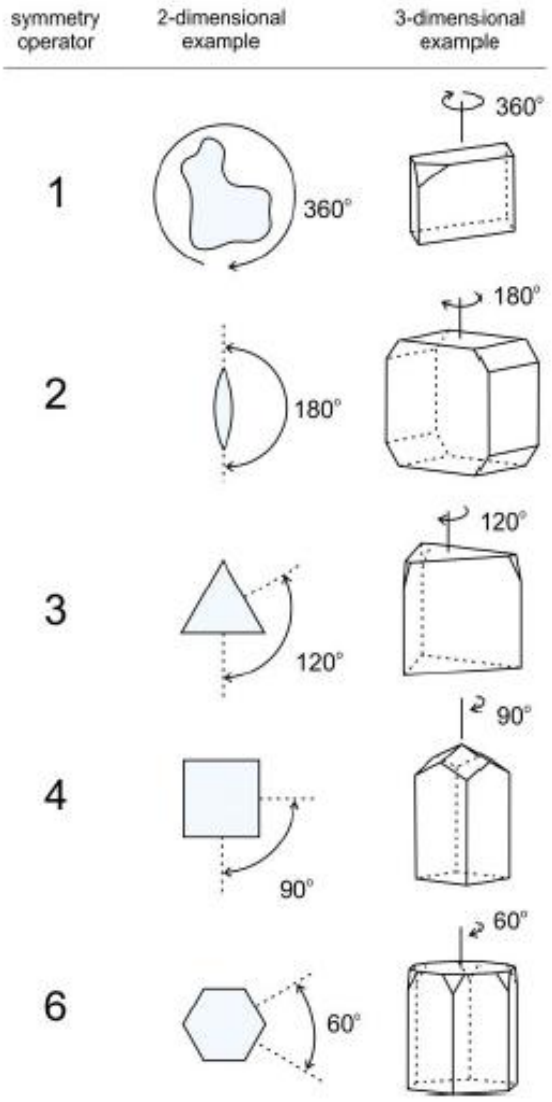


Left: *Polyhedron showing a two-fold rotation axis (**2**) passing through the centers of the top and bottom edges*
Right: *Polyhedron showing a reflection plane (**m**) that relates (as a mirror does) the top to the bottom*

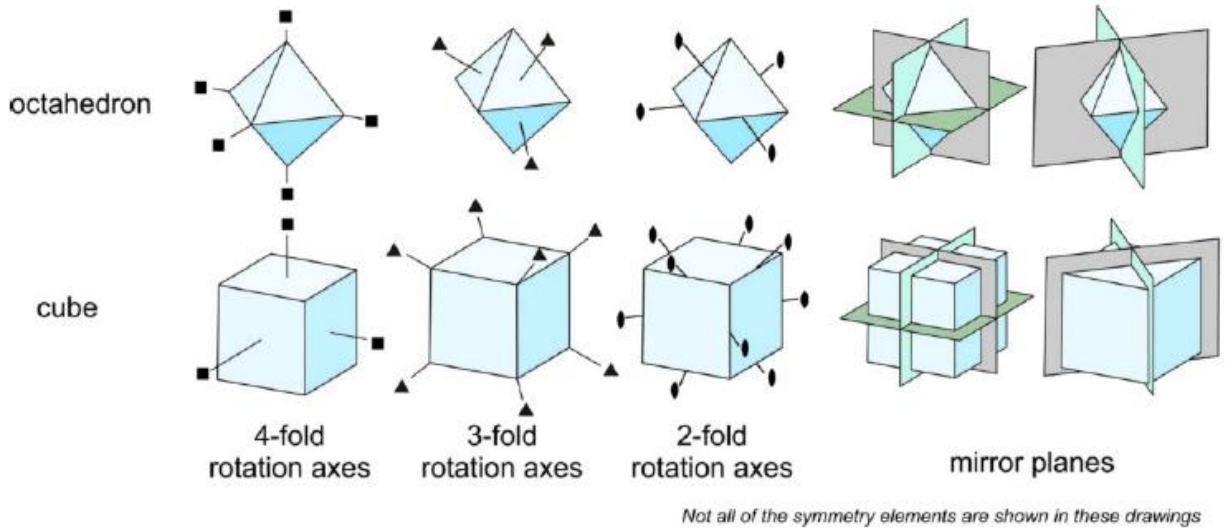
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Symmetry Elements

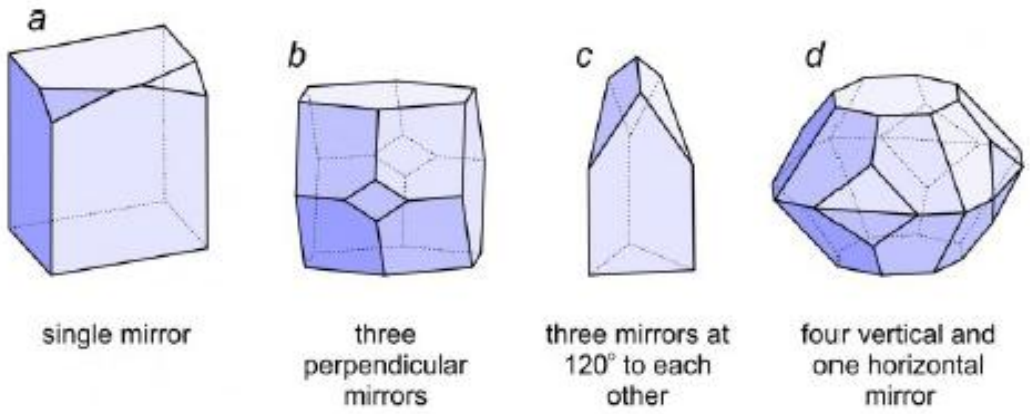
Rotational symmetry



Symmetry of an octahedron and of a cube



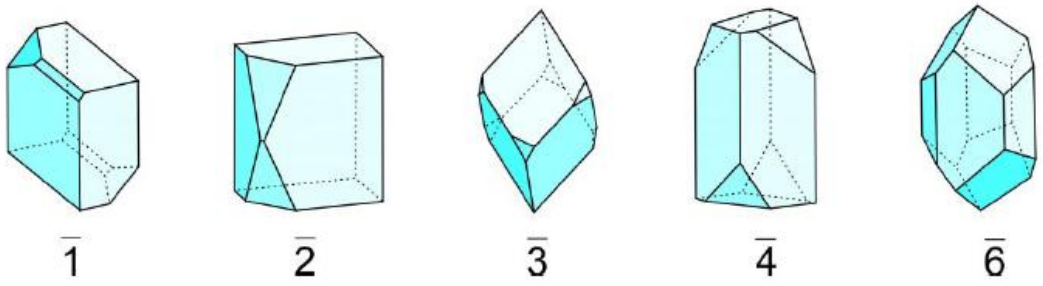
Crystals with mirror planes



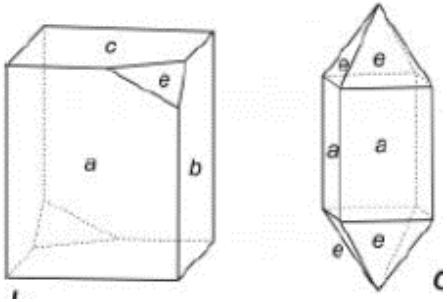
https://opengeology.org/Mineralogy/10-crystal-morphology-and-symmetry/

Symmetry Elements

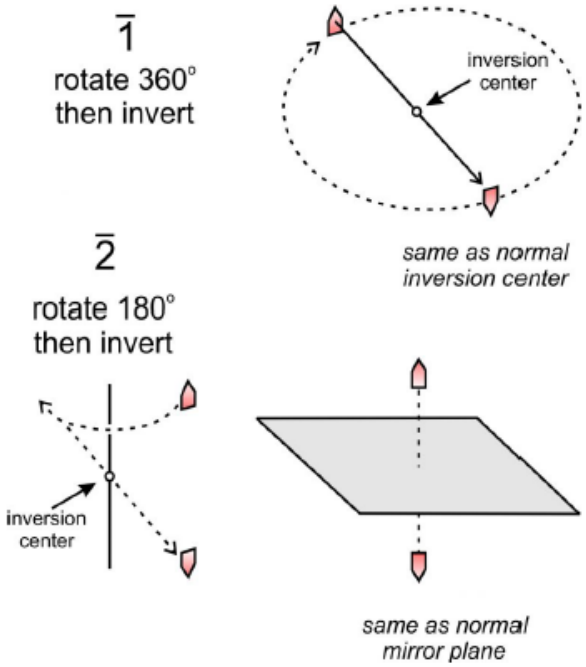
Crystals with rotoinversion symmetry



Inversion centers

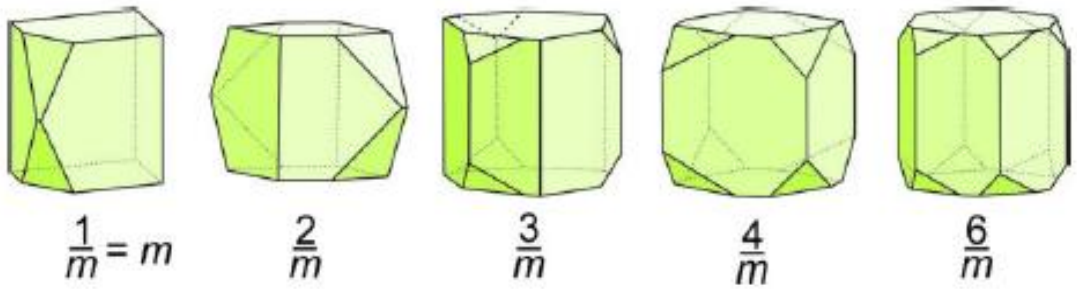


$\bar{1}$ and $\bar{2}$ rotoinversion operations



The crystal on left has an inversion center and no other symmetry; while, the crystal on the right contains an inversion center, but other symmetry is present. 90º rotation relates the four a-faces. Inversion and rotation relate the eight e-faces. Additionally, many mirror planes are present.

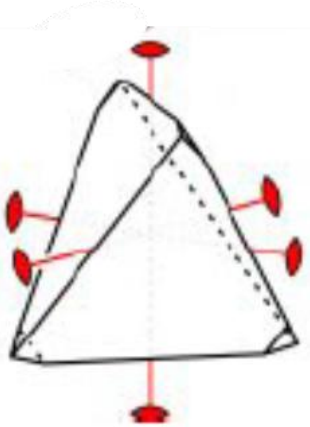
Crystals with rotation axes and perpendicular mirror planes



https://opengeology.org/Mineralogy/10-crystal-morphology-and-symmetry/

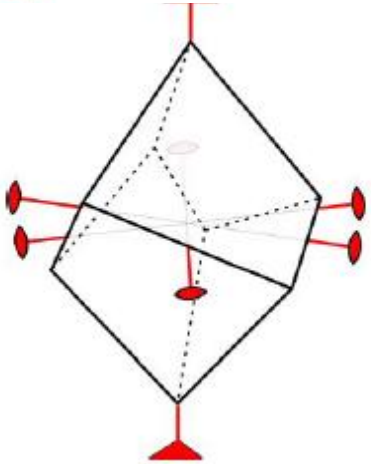
Symmetry Elements

- The symmetry operators can combine, so more than one can be present in a crystal
- But, the number of possible combinations is limited for two reasons
- First, some combinations lead to other symmetry
- Second, some combinations are contradictory and thus impossible
- We call distinct combinations of symmetry, such as 222, **point groups**
- There are 32 point groups



two perpendicular 2-fold axes require that a third be present

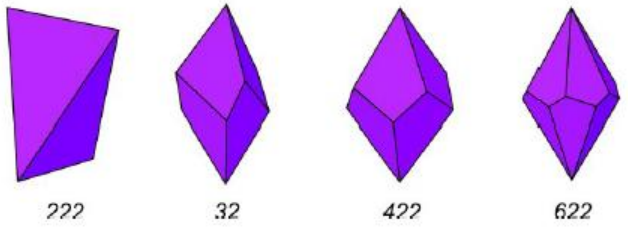
two disphenoids
222



two 2-fold axes that intersect at 60° require the presence of a third 2-fold axis and of a 3-fold axis

trapezohedron
32

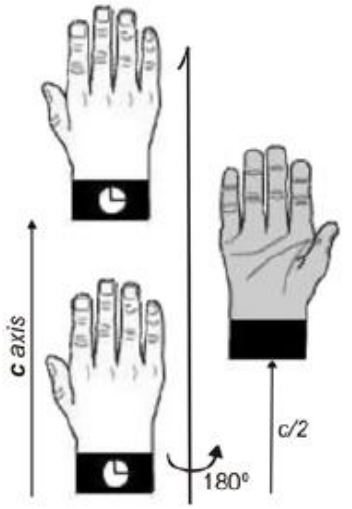
Crystals belonging to point groups 222, 32, 422, and 622



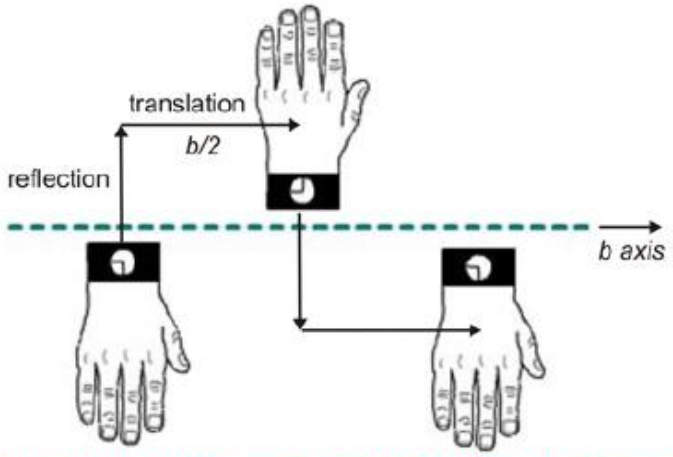
https://opengeology.org/Mineralogy/10-crystal-morphology-and-symmetry/

Symmetry Elements

- Out of 32 point groups, 11 contain center of symmetry operator and comprise Laue groups
- Additional symmetry operators that involve translation along with rotation/reflection include **glide planes** and **screw axes** (not included in point groups but in space groups)
- There are a total of 230 space groups - 230 different ways to replicate a finite object (motif) in 3D space
- These 230 ways to repeat patterns in space, which are compatible with the 32 crystal classes and with the 14 Bravais lattices, are called **space groups**
- They represent the 230 different ways to fit the Bravais lattices to the symmetry of the objects



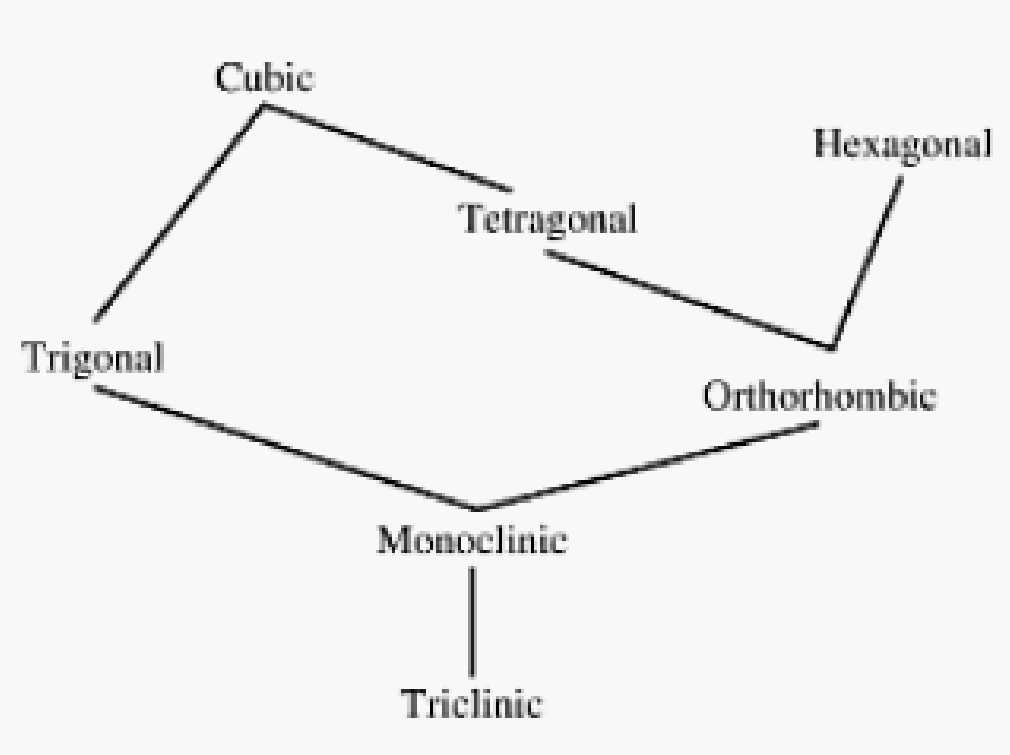
Twofold screw axis. A screw axis consists of a rotation followed by a translation



Glide plane. A glide plane consists of a reflection followed by a translation

https://www.xtal.iqfr.csic.es/Cristalografia/parte_03-en.html

Symmetry of 7 crystal systems



7 crystal systems ranked according to symmetry
Cubic has highest symmetry and Triclinic has least symmetry

Minimum Symmetry Elements of 7 crystal systems

System name	Minimum symmetry	Conventional unit cell
Cubic	four triads – 3 or $\bar{3}$ (// space diagonals of cube)	a = b = c; α = β = γ = 90°
Tetragonal	one tetrad – 4 or $\bar{4}$ (//Z)	a = b ≠ c; α = β = γ = 90°
Orthorhombic	three mutually ⊥ diads – 2 or m (//X, Y and Z)	a ≠ b ≠ c; α = β = γ = 90°
Rhombohedral/ Trigonal	one triad – 3 or $\bar{3}$ (typically //Z)	a = b = c; α = β = γ ≠ 90°
Hexagonal	one hexad – 6 or $\bar{6}$ (//Z)	a = b ≠ c; α = β = 90°; γ = 120°
Monoclinic	one diad – 2 or m (//Y)	a ≠ b ≠ c; α = γ = 90° ≠ β
Triclinic	1 or $\bar{1}$	a ≠ b ≠ c; α ≠ β ≠ γ ≠ 90°

Interplanar spacing

Cubic:
$$\frac{1}{d^2} = \frac{h^2 + k^2 + l^2}{a^2}$$

Tetragonal:
$$\frac{1}{d^2} = \frac{h^2 + k^2}{a^2} + \frac{l^2}{c^2}$$

Hexagonal:
$$\frac{1}{d^2} = \frac{4}{3}\left(\frac{h^2 + hk + k^2}{a^2}\right) + \frac{l^2}{c^2}$$

Rhombohedral:
$$\frac{1}{d^2} = \frac{(h^2 + k^2 + l^2)\sin^2\alpha + 2(hk + kl + hl)(\cos^2\alpha - \cos\alpha)}{a^2(1 - 3\cos^2\alpha + 2\cos^3\alpha)}$$

Orthorhombic:
$$\frac{1}{d^2} = \frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2}$$

Monoclinic:
$$\frac{1}{d^2} = \frac{1}{\sin^2\beta}\left(\frac{h^2}{a^2} + \frac{k^2\sin^2\beta}{b^2} + \frac{l^2}{c^2} - \frac{2hl\cos\beta}{ac}\right)$$

Triclinic:
$$\frac{1}{d^2} = \frac{1}{V^2}\left(S_{11}h^2 + S_{22}k^2 + S_{33}l^2 + 2S_{12}hk + 2S_{23}kl + 2S_{13}hl\right)$$

In the equation for triclinic crystals

$$\begin{aligned}
V &= \text{volume of unit cell (see below)},\\
S_{11} &= b^2c^2\sin^2\alpha,\\
S_{22} &= a^2c^2\sin^2\beta,\\
S_{33} &= a^2b^2\sin^2\gamma,\\
S_{12} &= abc^2(\cos\alpha\cos\beta - \cos\gamma),\\
S_{23} &= a^2bc(\cos\beta\cos\gamma - \cos\alpha),\\
S_{13} &= ab^2c(\cos\gamma\cos\alpha - \cos\beta).
\end{aligned}$$

Volume of unit cell

Cubic: $$V = a^3$$

Tetragonal: $$V = a^2 c$$

Hexagonal: $$V = \frac{\sqrt{3}\, a^2 c}{2} = 0.866 a^2 c$$

Rhombohedral: $$V = a^3 \sqrt{1 - 3\cos^2\alpha + 2\cos^3\alpha}$$

Orthorhombic: $$V = abc$$

Monoclinic: $$V = abc \sin\beta$$

Triclinic: $$V = abc \sqrt{1 - \cos^2\alpha - \cos^2\beta - \cos^2\gamma + 2\cos\alpha\cos\beta\cos\gamma}$$

Interplanar angles

Cubic:

$$\cos\phi = \frac{h_1h_2 + k_1k_2 + l_1l_2}{\sqrt{(h_1^2 + k_1^2 + l_1^2)(h_2^2 + k_2^2 + l_2^2)}}$$

Tetragonal:

$$\cos\phi = \frac{\dfrac{h_1h_2 + k_1k_2}{a^2} + \dfrac{l_1l_2}{c^2}}{\sqrt{\left(\dfrac{h_1^2 + k_1^2}{a^2} + \dfrac{l_1^2}{c^2}\right)\left(\dfrac{h_2^2 + k_2^2}{a^2} + \dfrac{l_2^2}{c^2}\right)}}$$

Hexagonal:

$$\cos\phi = \frac{h_1h_2 + k_1k_2 + \dfrac{1}{2}(h_1k_2 + h_2k_1) + \dfrac{3a^2}{4c^2}l_1l_2}{\sqrt{\left(h_1^2 + k_1^2 + h_1k_1 + \dfrac{3a^2}{4c^2}l_1^2\right)\left(h_2^2 + k_2^2 + h_2k_2 + \dfrac{3a^2}{4c^2}l_2^2\right)}}$$

Rhombohedral:

$$\cos\phi = \frac{a^4d_1d_2}{V^2}[\sin^2\alpha(h_1h_2 + k_1k_2 + l_1l_2) + (\cos^2\alpha - \cos\alpha)(k_1l_2 + k_2l_1 + l_1h_2 + l_2h_1 + h_1k_2 + h_2k_1)]$$

Interplanar angles

Orthorhombic:

$$\cos\phi = \frac{\dfrac{h_1h_2}{a^2} + \dfrac{k_1k_2}{b^2} + \dfrac{l_1l_2}{c^2}}{\sqrt{\left(\dfrac{h_1^2}{a^2} + \dfrac{k_1^2}{b^2} + \dfrac{l_1^2}{c^2}\right)\left(\dfrac{h_2^2}{a^2} + \dfrac{k_2^2}{b^2} + \dfrac{l_2^2}{c^2}\right)}}$$

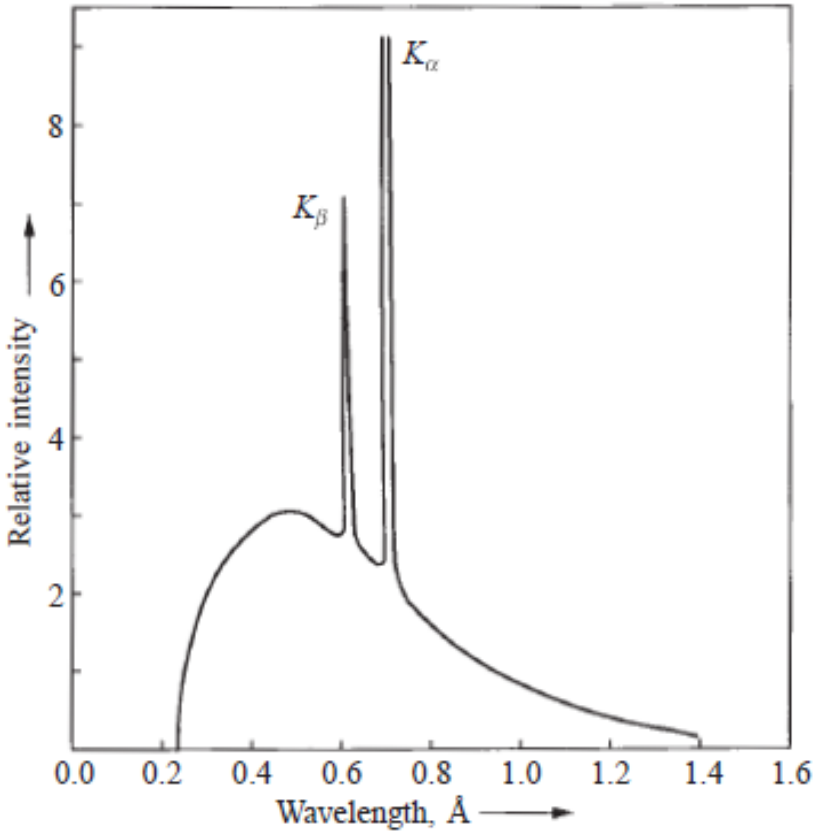
Monoclinic:

$$\cos\phi = \frac{d_1d_2}{\sin^2\beta}\left[\frac{h_1h_2}{a^2} + \frac{k_1k_2\sin^2\beta}{b^2} + \frac{l_1l_2}{c^2} - \frac{(l_1h_2 + l_2h_1)\cos\beta}{ac}\right]$$

Triclinic:

$$\cos\phi = \frac{d_1d_2}{V^2}[S_{11}h_1h_2 + S_{22}k_1k_2 + S_{33}l_1l_2 + S_{23}(k_1l_2 + k_2l_1) + S_{13}(l_1h_2 + l_2h_1) + S_{12}(h_1k_2 + h_2k_1)]$$

Crystal structure determination by XRD



The spectrum of x-rays emitted from a molybdenum target at 35 kV

Wavelengths of K_α Radiation for Typical Target Metals

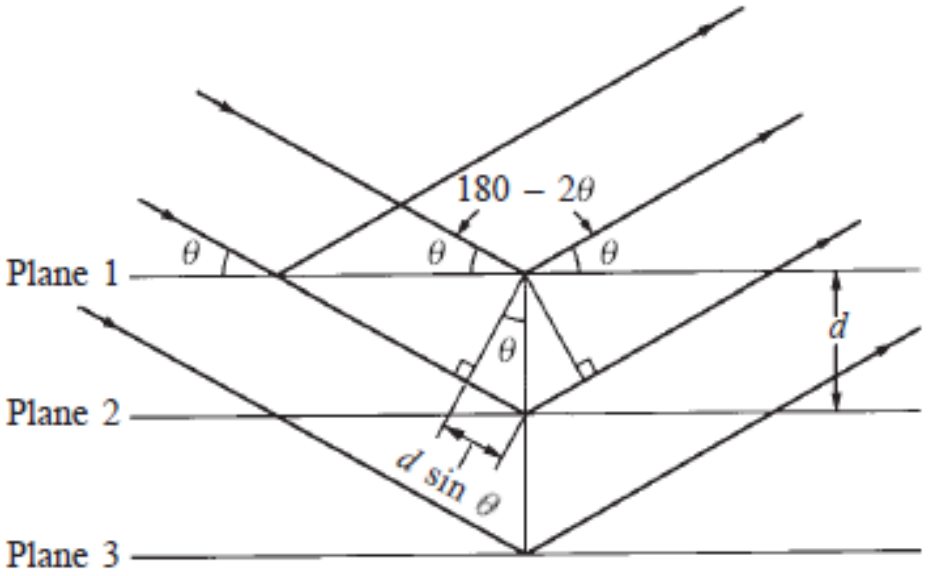
Target metal		Mo	Cu	Co	Fe	Cr
K_α wavelength,	Å	0.71	1.54	1.79	1.94	2.29
	nm	0.071	0.154	0.179	0.194	0.229

Crystal structure determination by XRD

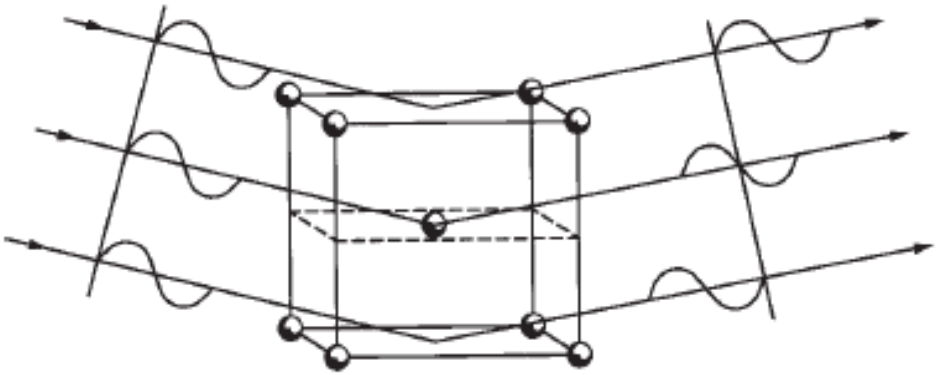
Bragg Law

	λ	θ
Laue method	Variable	Fixed
Rotating-crystal method	Fixed	Variable (in part)
Powder method	Fixed	Variable

$$n\lambda = 2d \sin \theta$$



Crystal structure determination by XRD



The (100) reflection is absent for a BCC crystal, as reflection from the corner atoms is exactly cancelled out by that from body centered atoms

Extinction Rules for Cubic Crystals

Crystal	Reflections are allowed
SC	for all values of $(h^2 + k^2 + l^2)$
BCC	for even values of $(h + k + l)$
FCC	when h, k and l are all odd or all even
DC	when h, k and l are all odd, or when all are even, $(h + k + l)$ should be divisible by four

Crystal structure determination by XRD

For example, if the crystal is cubic, then

$$\lambda = 2d \sin\theta$$

and

$$\frac{1}{d^2} = \frac{(h^2 + k^2 + l^2)}{a^2}.$$

Combining these equations, we have

$$\sin^2\theta = \frac{\lambda^2}{4a^2}(h^2 + k^2 + l^2).$$

Atomic scattering factor

$$f = \frac{\text{amplitude of the wave scattered by an atom}}{\text{amplitude of the wave scattered by one electron}}$$

$$e^{ix} = \cos x + i \sin x$$

$$Ae^{i\phi} = A\cos\phi + Ai\sin\phi$$

$$Ae^{i\phi} = fe^{2\pi i(hu+kv+lw)}$$

$$F = f_1 e^{2\pi i(hu_1+kv_1+lw_1)} + f_2 e^{2\pi i(hu_2+kv_2+lw_2)} + f_3 e^{2\pi i(hu_3+kv_3+lw_3)} + \cdots$$

$$F_{hkl} = \sum_1^N f_n e^{2\pi i(hu_n+kv_n+lw_n)}$$

Structure factor

$$|F| = \frac{\text{amplitude of the wave scattered by all the atoms of a unit cell}}{\text{amplitude of the wave scattered by one electron}}$$

Crystal structure determination by XRD

Bravais lattice	Reflections present	Reflections absent
Simple	all	none
Base-centered	h and k unmixed*	h and k mixed*
Body-centered	$(h + k + l)$ even	$(h + k + l)$ odd
Face-centered	h, k, and l unmixed	h, k, and l mixed

Close packed hexagonal

$h + 2k$	l	$\lvert F\rvert^2$
$3n$	odd	0
$3n$	even	$4f^2$
$3n \pm 1$	odd	$3f^2$
$3n \pm 1$	even	f^2

References

- Materials Science and Engineering, A First Course by V. Raghavan
- Fundamentals of Materials Science and Engineering, W.D. Callister
- Structure of Materials by Marc De Graef and Michael E. McHenry
- Elements of X-Ray Diffraction by B.D. Cullity